

Domain of Validity of the Emergent Gravity Program: AdS Results, de Sitter Obstructions, and the Remaining Open Question

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Abstract

Papers I–VIII of the Emergent Gravity Program establish, inside holographic conformal field theories with AdS asymptotics, that the finite-ball entanglement first law is equivalent to the linearized Einstein–Cartan equations (metric sector unconditional, axial sector after a finite dictionary improvement) and that the Hehl–Datta coefficient is fixed by pure information. The present note records the final open question of the program: the domain of validity of these results beyond AdS. Every **[P]**-grade derivation relies on the Casini–Huerta–Myers modular Hamiltonian for balls, a global conformal structure, and a clean asymptotic boundary. None of these ingredients is known to exist in the same form in de Sitter space or in a generic cosmology. The AdS results stand independently; their extrapolation to cosmology remains **[O]**. Labels follow the standing convention of the program.

Why this work is required. A claim that “the gravitational sector of known physics is entanglement bookkeeping” is incomplete until its domain is stated with the same precision as its derivations. The AdS sector is now closed. The cosmological sector is not. This note isolates the precise obstructions and the minimal requirements for a future derivation, so that the public record of the program is accurate and no stronger claim is made than the evidence supports.

Epistemic Conventions

Every material claim is tagged **[P]** (proved/derived within a stated framework), **[C]** (conjectured with support) or **[O]** (open). Numerical agreement is not proof. Consensus is not proof.

1 What Has Been Established

Inside the holographic AdS regime the following statements are certified:

1. The first law of entanglement on all balls is equivalent to the linearized Einstein equations about AdS (Faulkner–Guica–Hartman–Myers–Van Raamsdonk). **[P]**
2. The modular perturbation kernel is exact and machine-verified. **[P]**

3. The metric sector of the torsionful first law for finite balls is unconditional by the conformal selection rule. [P]
4. The axial channel produces a universal kinematic residue I_B that is non-vanishing for finite regions and absorbable into a finite dictionary improvement. [P]
5. The sign of the Hehl–Datta coefficient follows from relative-entropy positivity. [P]
6. The magnitude is fixed both by rigidity under program normalizations and by pure-information Fisher–canonical matching (null-plane limit + absorbability). [P]

Collectively these results establish the Einstein–Cartan certification of the New Standard Model from entanglement bookkeeping inside the AdS domain.

2 The Final Open Question

Almost every step above uses one or more of the following ingredients that are special to AdS/CFT:

- the Casini–Huerta–Myers modular Hamiltonian for ball-shaped regions, which relies on a conformal map to a hyperbolic geometry;
- a global conformal structure on the boundary;
- a clean asymptotic boundary with a well-defined holographic dictionary;
- the existence of a Ryu–Takayanagi surface and a bulk entanglement wedge with a Killing horizon dual to the modular flow.

None of these is known to possess a direct counterpart of equal rigor in de Sitter space. In de Sitter:

- there is no conformal boundary of the same type;
- static-patch modular Hamiltonians exist but are not given by a CHM-type local integral of the stress tensor in the same way;
- the absence of a global timelike Killing vector outside the static patch complicates the notion of a first law for arbitrary ball-like regions;
- no generally accepted microscopic dual exists that would supply an analogue of the FGHMV equivalence.

Cosmology in this program is a *metric* question. Torsion is algebraic, vacuum-vanishing, and does not propagate: it is not a cosmological limitation and is not part of the domain bar. The open claim is that the same bookkeeping produces Einstein gravity with Λ in a cosmological setting. That claim is at present [O] at the FGHMV standard (sign and isometries; Paper 17). The spinless vacuum of the metric sector, when $\Lambda > 0$, is de Sitter; that is Einstein+ Λ , not a holographic selection of cosmology.

3 Minimal Requirements for Closure

A derivation that would close the *metric* domain question at the standard of the AdS results must:

1. identify a family of regions in de Sitter (or a quasi-local cosmological horizon) whose modular Hamiltonians are under analytic control;
2. establish a first-law identity relating the variation of entanglement entropy to a bulk Noether charge or canonical energy, with the sign derived, not copied from AdS;
3. show that the resulting constraints are equivalent to the linearized Einstein equation with Λ , $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$.

Torsion and Hehl–Datta are not items on this list. Until such a derivation exists, the holographic extrapolation remains open. The vacuum solution itself does not wait on that derivation.

4 Certified Status of the Program

- AdS / holographic regime with ball regions: closed. [P]
- de Sitter / generic cosmological regime: open. [O]
- Non-linear / non-perturbative completion: open. [O]
- UV completion of the effective Lagrangian: open by construction (the program is a theory of everything we know, not a theory of everything). [O]

The AdS results stand on their own. No claim of the program that has been labeled [P] is weakened by the domain limitation. The limitation is simply recorded with the same precision that the derivations themselves were recorded.

Acknowledgments

This note completes the public accounting of open questions in the Emergent Gravity Program as of 14 August 2026. All epistemic classifications are the responsibility of the author.

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